

1 What are ensemble learning methods and what are they effective?

- Ensemble learning combines multiple machine learning models to make better predictions than single model. It improves accuracy, reduces errors, and makes the models more robust by combining the strengths.

2 Explain the difference between Bagging, Boosting and voting.

- Bagging: Training multiple models independently on different random subsets of data and combines their predictions.
- Boosting: training models one after another where each new model corrects the mistakes of the previous one.

voting: combines predictions from different trained models and selects the final output using majority voting (classification) or averages (regression).

3 What is Bias-variance Trade-off and how do ensemble methods address it?

- The Bias-variance Trade-off balances underfitting (high bias) and overfitting.

Bagging reduces variance, Boosting reduces bias and together they help improve the model's overall performance.

4 Explain votes classifier and Stack's ensemble.

- A votes classifier combines predictions from multiple models using majority votes or probability averages.

5 Compare Adaboost, Gradient Boosting, XGBoost, and LightGBM.

⇒ i) Adaboost = Focuses more on misclassified samples by increasing their weights in each iteration.

ii) Gradient Boosting = Build models sequentially by minimizing prediction error using gradient descent.

iii) XGBoost = An optimized regression of gradient boosting with regularization, faster training & better performance.

iv) LightGBM : A Fast and memory efficient Gradient Boosting framework that uses leaf-wise tree growth for large datasets.